

Creator Collaboration Hub (Cosmo)

1. Purpose & Use of Project

Creator Collaboration Hub (Cosmo) is a cinematic, premium platform designed for elite designers, developers, and creative visionaries to connect, showcase their work, and build breakthrough projects together.

Unlike traditional job boards that separate "employers" and "applicants," this platform uses a **unified creator ecosystem**. Every user is a "Creator" who can seamlessly switch contexts—posting their own visionary projects one minute, and applying to collaborate on someone else's project the next.

Key Objectives:

- **Centralized Profiles:** A single, robust creator profile showcasing skills, bio, downloadable PDF resumes, and a visual portfolio.
- **Project Discovery:** A dynamic ecosystem where users can explore active projects, filter by status, and discover trending skills.
- **Streamlined Collaboration:** Project owners can easily review applicants, accept/reject pitches, and notify collaborators in real-time.
- **Premium User Experience:** A glassmorphic, dark-themed UI enhanced by smooth Framer Motion micro-animations to provide a "cinematic" and highly engaging user experience.

2. Core Workflows

A. Authentication & Identity Flow

1. **Onboarding:** A user registers with an email, username, and password.
2. **Stateless Auth:** The Spring Boot backend authenticates the user and issues a JWT (JSON Web Token).
3. **Session:** The React frontend stores the JWT and attaches it to the Authorization header for all subsequent protected API calls.

B. Creator Profile & Portfolio Workflow

1. **Profile Initialization:** The user populates their profile (Bio, Links, Experience Level, Skills).
2. **Asset Uploads:** The user can upload an Avatar or a PDF Resume. The backend securely pushes these files to **Cloudinary** and stores the resulting secure URLs in the MySQL database.
3. **Portfolio Management:** Creators can upload case studies or past work (Thumbnails, Titles, Descriptions, Tech Stack) to their portfolio, which dynamically renders in a visually striking grid on their public profile.

C. Project Collaboration Lifecycle

1. **Project Creation (Owner):** A creator posts a new collaboration project, defining required roles, project scope, and a banner image.
2. **Discovery & Application (Collaborator):** Other creators browse the ecosystem, find the project, and submit an application containing a pitch message, portfolio link, and their desired role.
3. **Review & Action (Owner):** The project owner receives the application in their Dashboard. They can review the applicant's profile/resume and choose to **Accept** or **Reject** the application.
4. **Notification:** The applicant's dashboard instantly reflects the updated status of their application.

3. Technology Stack

Frontend (Client-Side)

- **Framework:** React (bootstrapped with Vite for ultra-fast HMR and optimized builds).
- **Styling:** Custom CSS combined with Tailwind utility patterns to achieve a high-end **Glassmorphism** aesthetic (frosted glass panels, ambient glows, blur effects).
- **Motion & Animation:** framer-motion is used extensively for cinematic page transitions, staggered list animations, and interactive hover states.
- **Icons:** lucide-react for clean, modern, and lightweight SVG icons.
- **Routing & State:** react-router-dom for SPA routing, and React Context (AuthContext) for global state management of the user's session.

Backend (Server-Side)

- **Framework:** Spring Boot 3 (Java 17).
- **Security:** Spring Security with custom stateless JWT Filters for robust endpoint protection.

- **API Design:** RESTful architecture, heavily utilizing DTOs (Data Transfer Objects) to decouple the database entities from the JSON payloads sent to the client.
- **Documentation:** `springdoc-openapi` (Swagger UI) for automated API contract generation.

Database & Storage

- **Relational Database:** MySQL 8.0.
- **ORM:** Spring Data JPA / Hibernate for object-relational mapping, automatic schema generation, and query building.
- **Cloud Storage:** **Cloudinary** SDK is integrated to handle all multipart file uploads (Profile Images, Project Banners, and PDF Resumes). It serves files securely via CDN, offloading storage overhead from the local server.